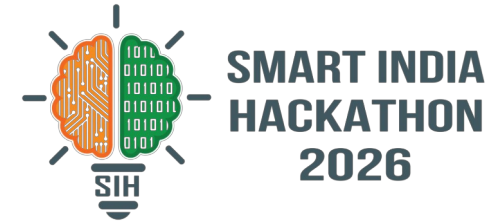


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- **Problem Statement ID – 26106**
- **Problem Statement Title - AI-powered Email Threat Detection, Geolocation and Forensic Intelligence Platform**
- **Theme- Blockchain & Cybersecurity**
- **PS Category - Software**
- **Team ID -**
- **Team Name (Registered on portal) - The Mentalists**





IDEA :- Automated Email Forensics & Threat Intelligence SOC Platform



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Problem fit • proposed solution •
innovation

THE PROBLEM

Modern cyber threats exploit email vectors through complex phishing, spoofed headers, and multi-hop relay obfuscation. Security Operations Center (SOC) analysts face critical bottlenecks:

- Manual .eml/.msg header analysis is slow and error-prone
- Limited SMTP-hop and GeoIP visibility
- No tamper-evident evidence ledger for legal use

PROPOSED SOLUTION

An integrated, automated Email Forensics & Threat Intelligence Platform that ingests raw email artifacts, extracts deep metadata, and generates court-admissible dossiers:

- Automated header extraction and SPF / DKIM / DMARC authentication scoring.
- Interactive SMTP Routing Topology Graph and GeoIP map visualization.
- Cryptographic hash chaining to guarantee evidence integrity for court submission.

WHY IT IS DIFFERENT

- SHA-256 hash-chained chain of custody
- NLP urgency + GeoIP intelligence
- One-click standardized forensic PDF dossier

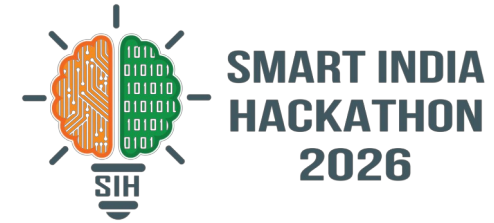


Automated evidence journey — from raw email artifact to verifiable forensic output



Architecture + implementation
pipeline

TECHNICAL APPROACH



FRONTEND

Next.js + TypeScript + Tailwind • SOC Dashboard • React Flow topology • Leaflet GeolP map

BACKEND

FastAPI + Python • Email Parser • SPF/DKIM/DMARC verification • NLP urgency scoring • Threat scoring

EVIDENCE & INTELLIGENCE

Email metadata • SMTP hop data • GeolP intelligence • SHA-256 audit ledger • Court-ready PDF dossier

Implementation Process & Pipeline :-

01

File Ingestion

Upload .eml / .msg raw emails via React Dropzone to POST /api/analyze API.

02

Header & NLP Parsing

Extract Received headers, SPF/DKIM validation, & NLP urgency scoring

03

Topology & GeolP Map

Reconstruct SMTP hop network graph and pin origin coordinates on Leaflet map.

04

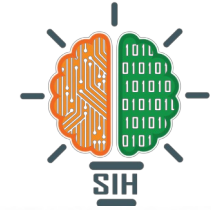
Ledger & PDF Export

Hash-chained audit log creation & 1-click generation of court PDF report.



Risk → mitigation → implementation
confidence

FEASIBILITY AND VIABILITY



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IMPACT AND BENEFITS



Key standards, protocols and source material

01



SMTP AUTHENTICATION

Email authentication & SMTP security

02



MAGIC BYTES & FILE VALIDATION

File signature /upload validation

03



SENTENCE TRANSFORMERS

Semantic text similarity

04



INFORMATION ENTROPY

Entropy - based analysis

05



LEVENSHTEIN DISTANCE

String similarity /typo detection

06



spaCy ENGLISH MODELS

NLP & language processing